

Yasser Taha, Machine Learning R&D

Berlin, Deutschland, 015731033474, ysr.mahmoud@gmail.com

PROFILE

Ω Working in a dynamic and multidisciplinary scientific environment that fosters seamless information exchange and diverse perspectives Ω Successfully contributed to a range of interdisciplinary machine learning (ML) projects, spanning Autonomous Systems, Pharmaceutical Recommender Systems, and Early Diagnostics Ω Demonstrated a track record of propelling projects from initial research stages to tangible and impactful developments Ω Cultivated a rich perspective by residing in four different countries, embracing diverse norms and enriching life experiences.

EMPLOYMENT HISTORY

Oct 2024 — Present	PhD Researcher, Robert Koch Institute Berlin
	Leading a project to develop an AI-driven tool for identifying infected cell cultures, including those caused by viruses that produce limited or no visible CPE. This tool has the potential to increase virus isolation rates, detect unknown viruses, and speed up the isolation process. The methodology can be applied to various research areas in virus detection using imaging techniques. <i>Methods: ViLSTM, Vision Transformer, Conformal prediction</i>
Apr 2021 — Jul 2024	Machine Learning Researcher, HB-DX Diagnostics Heidelberg
	Worked on Discovery of RNA biomarkers for early lung cancer detection from patients' blood biopsy resulting in 0.76 (stage I) to 0.81 (Stage IV) AUCROC. Conceived an explainable " <u>TransfoRNA</u> " ML <u>framework</u> for mapping small non-coding RNA NGS fragments back to their parental precursor, shedding light on their functional role and involvement in disease mechanisms. Developed a statistical data pipeline facilitating detailed comparison between fragment RPM counts (NGS) vs CPM (dPCR). <i>Methods: Transformers, Explainability, Contrastive Learning, Multimodality, Visualization.</i>
Oct 2020 — Apr 2021	Machine Learning R&D Engineer, Cognigy Düsseldorf
	Developed ConveRT's state of the art language model and trained it on a response selection task based on Reddit dataset. The model was then utilized for intent detection. <i>Methods: NLP, Transformers, Unsupervised Learning, Generative Models</i>
Oct 2019 — Sept 2020	Machine Learning Researcher, STTech (TUM) Munich
	Exploit the cross-selling relations between different pharmaceutical products as well as product ingredients to provide better recommendations. Devised a novel approach to re-rank edge weights in the cross-selling graph to overcome popularity bias in retail. The project is in production at Pharmatechnik. <i>Methods: Bayesian Statistics, GNNs, Autoencoders, Contrastive Learning, Recommender Systems</i>
Aug 2018 — Sept 2019	Research Assistant, Institute of Machine Learning, JKU Linz
	Supervised by Professor Sepp Hochreiter Researched and implemented the state-of-the-art approach for Unsupervised Depth Estimation from monocular sequence of images. Reproduced the state-of-the-art results and added two novel losses to the framework that led to better evaluation on the Kitti dataset. <i>Methods: Computer Vision, Flow Nets, Unsupervised Learning, Convolutional Autoencoders</i>
Jul 2016 — Aug 2017	Algorithm Engineer, Autonomous Vehicles at Valeo Cairo
	Implemented online external calibration for LiDAR sensor allowing vehicles to adapt more robustly to changes in sensor orientation. Designed and implemented ground detection algorithm from laser 3D point cloud. Both algorithms are currently deployed in production. <i>Methods: Computational Geometry, 3D Point Cloud, Kalman Filters</i>
May 2014 — Sept 2015	Teaching Assistant, German University In Cairo Cairo
	Teaching: Sensor technology, Pneumatics and PLC. Develop projects through research and implementation. Aid students performing their bachelor projects.

PUBLICATIONS

Dec 2025	Drainage: A Unifying Framework for Addressing Class Uncertainty	CVPR 2026
	First Author, DOI: https://doi.org/10.48550/arXiv.2512.03182	
Jul 2024	TransfoRNA: Navigating the Uncertainties of Small RNA Annotation with an Adaptive Machine Learning Strategy	Biorxiv
	First Author, DOI: https://doi.org/10.1101/2024.06.19.599329	
	Website: www.transforna.com	
Jun 2023	Early Detection of Lung Cancer using small RNAs	Journal of Thoracic Oncology
	DOI: 10.1016/j.jtho.2023.07.005	
Nov 2020	Graph CNN for a Pharmacy Cross-Selling Recommender System	Information
	First co-author, DOI: https://doi.org/10.3390/info11110525	

EDUCATION AND TRAINING

May 2025	AI in Public Health Research Symposium	Berlin
	Poster: AI-Driven Classification of Virus Infections via Cytopathic Effect Imaging	
Apr 2024	AI and Biology, EMBL	Heidelberg
Jun 2023	ML for Healthcare and Biology, ELLIS Summer School	Manchester
Apr 2023	Causal Learning and Reasoning, Conference	Tübingen
Sept 2017 — Jul 2018	MSc, Data Science, Johannes Kepler University	Linz
	Thesis Title: “Trajectory Prediction in the Realm of Sparse Data”	
	Full Scholarship Thesis GPA: 2.0	
Oct 2015 — Jul 2016	PgDip in Embedded Systems, Information Technology Institute	Cairo
	Full scholarship Funded by Valeo and IBM.	
Sept 2008 — Feb 2014	BSc, Mechatronics Engineering, German University in Cairo	Cairo
	Thesis Title: Four degrees of Freedom, automated, localized, and mapped Robot.	
	Thesis GPA: 1.0 Bachelor GPA: 2.0	

SKILLS

Unix	Supervised/Unsupervised Learning
Python, R, C	Computer Vision
PyTorch, TensorFlow	Language Model
Kubernetes, Docker	Geometry
Artificial Intelligence	small RNA
Bayesian Statistics	Computational Biology
Probabilistic models	Teaching
Statistics	Research
Recommender Systems	Deep Learning

LANGUAGES

German	C1	English	C2
Egyptian	Native speaker		